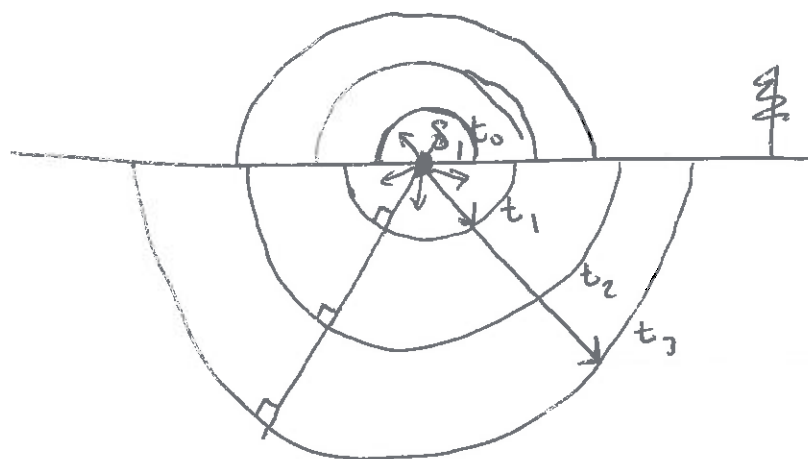
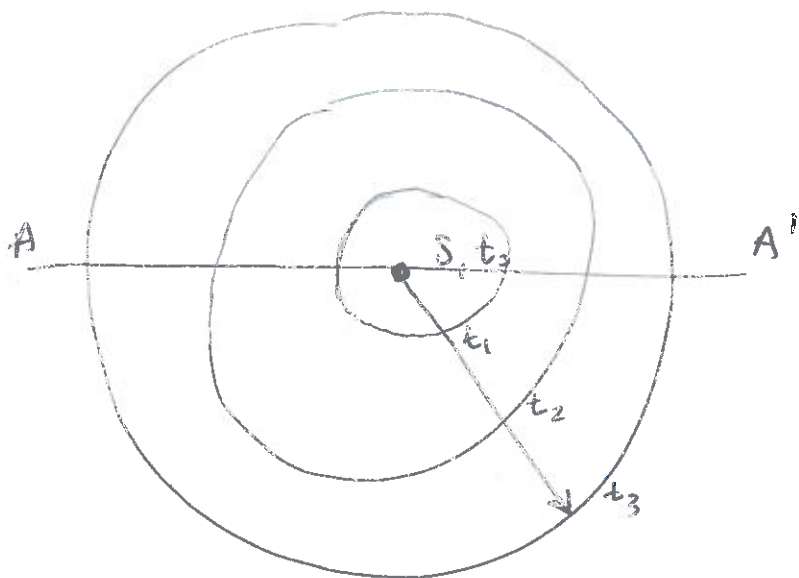
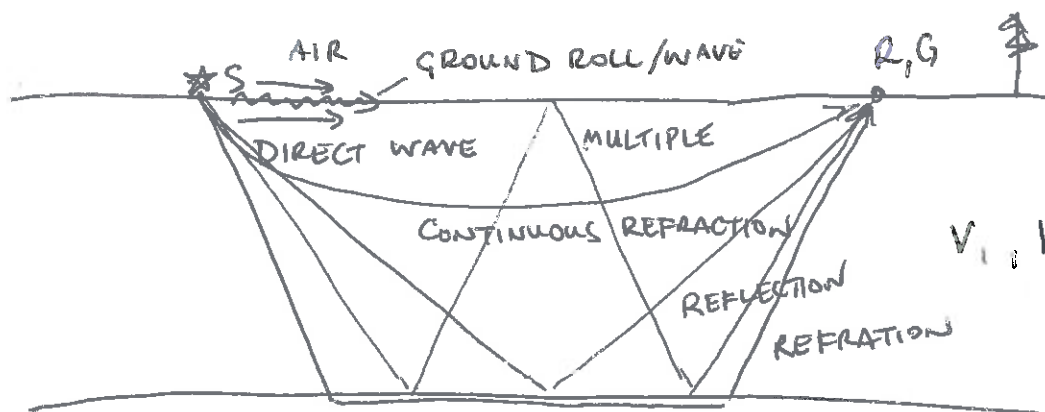


MAP VIEW
↑ N



$$\lambda_{\text{WAVEFRONT}} = t_{\text{ELAPSED}} v_{\text{MEDIUM}}$$



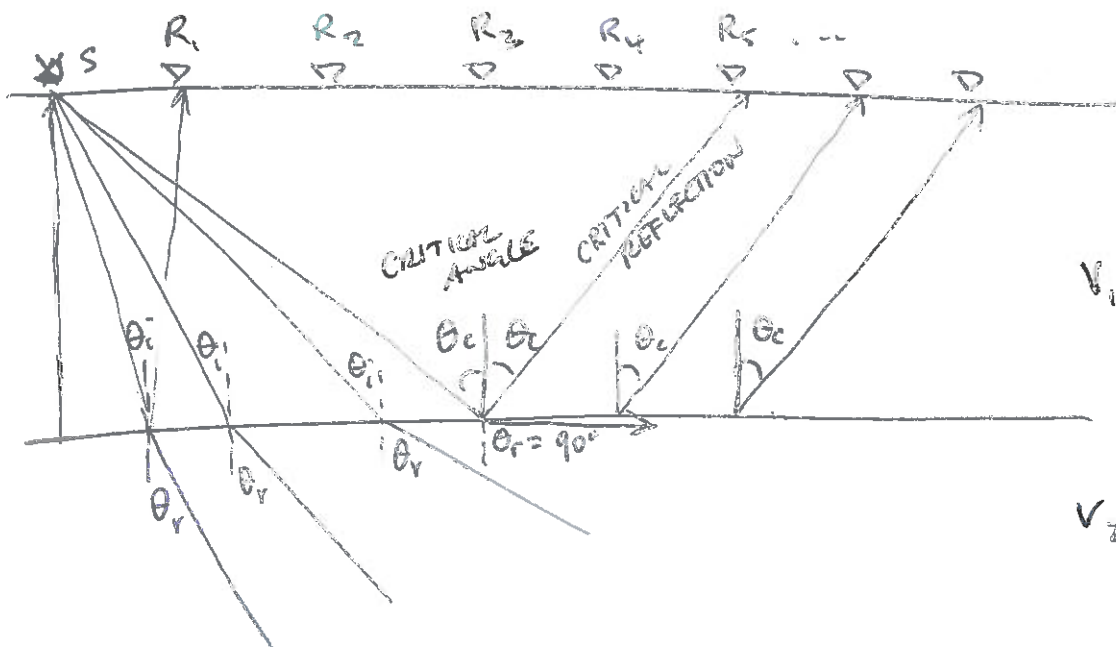
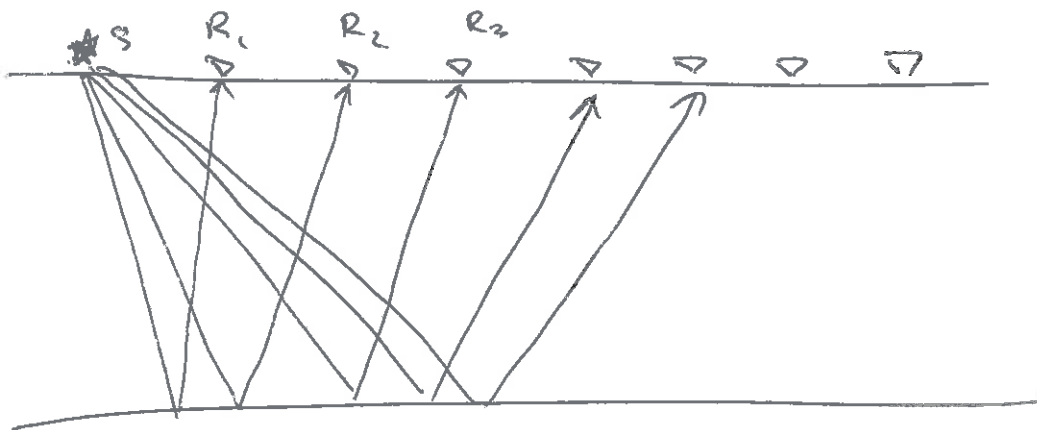
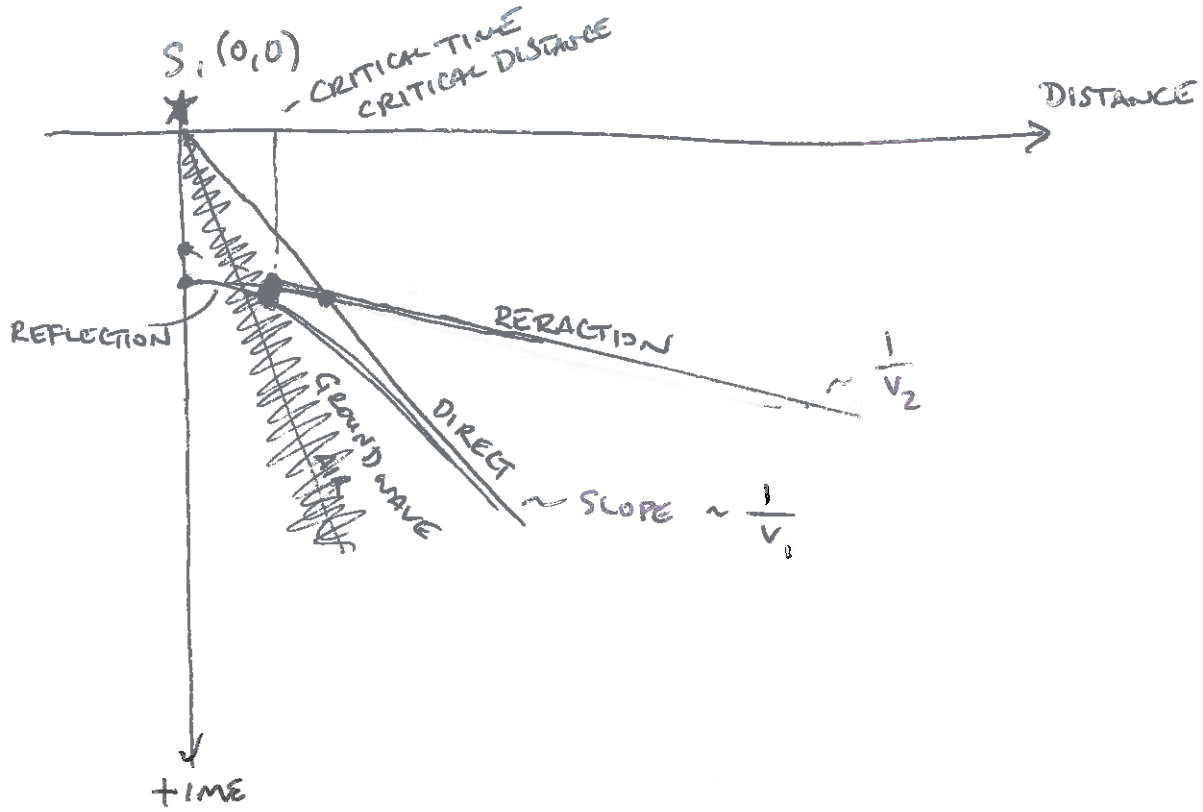
$$v_{\text{DIRECT}} > v_{\text{AIR}} > v_{\text{GROUND ROLL}}$$

v_1, h_1

$$v_2 > v_1$$

v_2

TRAVEL-TIME DIAGRAM



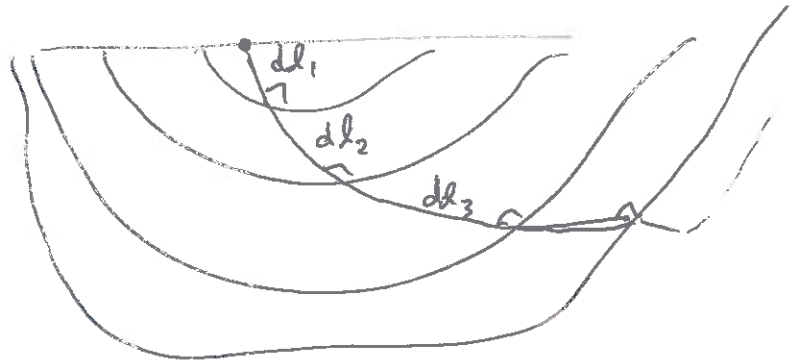
$t = \frac{l}{v}$

 travel time $\leftarrow$

 LENGTH OF PATH $\leftarrow l$

 VELOCITY $\leftarrow v$

$$t = \min \int_S^R \frac{dl}{v(x, y, z)}$$

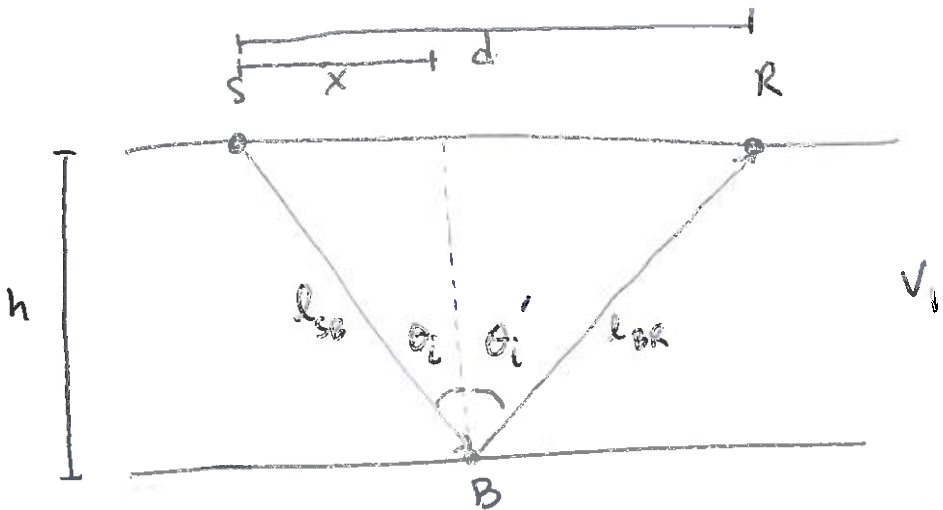


REFLECTION

ISOTROPIC

NO VELOCITY GRADIENTS

HORIZONTAL INTERFACES



$$t = \frac{l_{SB}}{v_1} + \frac{l_{BR}}{v_1}$$

$$l_{SB} = (x^2 + h^2)^{1/2}$$

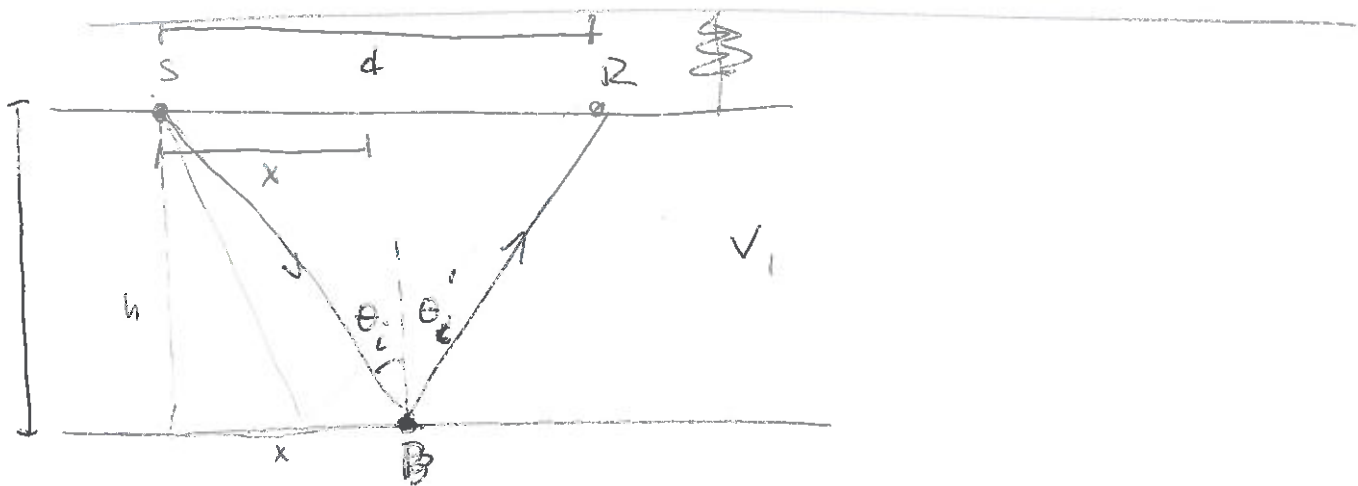
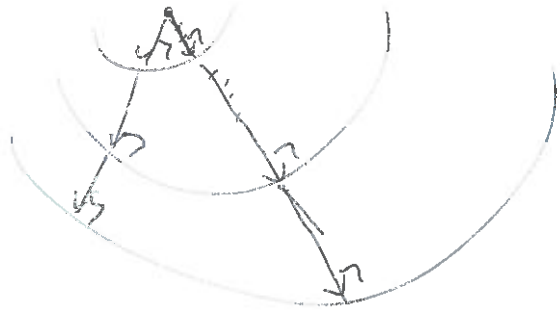
$$l_{BR} = ((d-x)^2 + h^2)^{1/2}$$

$$\min \left[t = \frac{(x^2 + h^2)^{1/2}}{v_1} + \frac{((d-x)^2 + h^2)^{1/2}}{v_1} \right]$$

$$\frac{dt}{dx} = 0 = \frac{1}{v_1} \left[\frac{x}{(x^2 + h^2)^{1/2}} - \frac{(d-x)}{((d-x)^2 + h^2)^{1/2}} \right]$$

$$t = \frac{l}{v}$$

$$t = \min \int_S^R \frac{dl}{v(x, y, z)}$$



$$t = t_{SB} + t_{BR}$$

$$t = \frac{l_{SB}}{v_1} + \frac{l_{BR}}{v_1}$$

TOTAL TRAVEL-TIME IS TIME OF EACH SEGMENT.

$$\min t = \frac{(x^2 + h^2)^{1/2}}{v_1} + \frac{((d-x)^2 + h^2)^{1/2}}{v_1}$$

TO FIND MINIMUM, TAKE DERIVATIVE, SET TO ZERO.

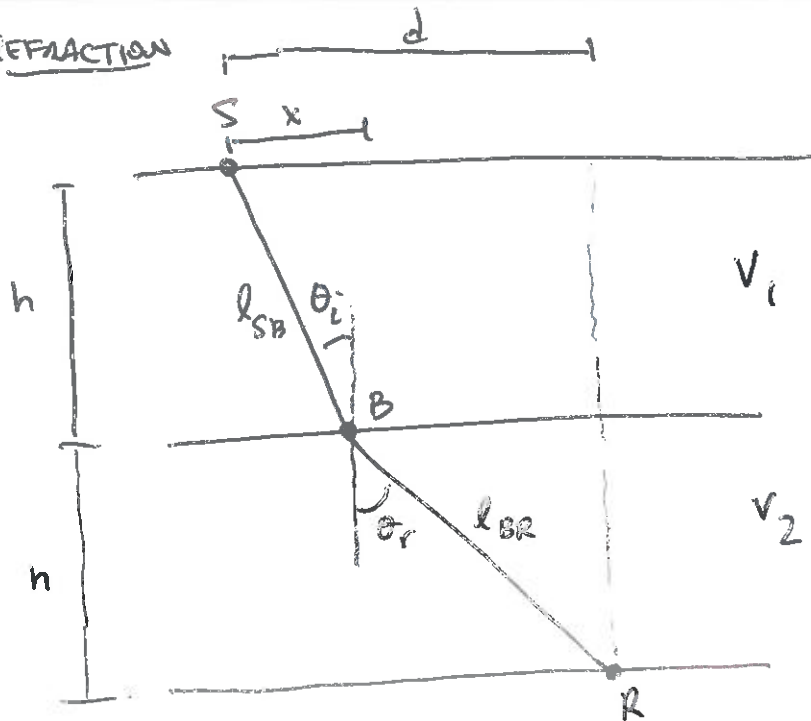
$$\frac{dt}{dx} = 0 = \frac{1}{v_1} \left[\frac{x}{(x^2 + h^2)^{1/2}} - \frac{(d-x)}{((d-x)^2 + h^2)^{1/2}} \right]$$

$$\frac{(d-x)}{((d-x)^2 + h^2)^{1/2}} = \frac{x}{(x^2 + h^2)^{1/2}}$$

$$\frac{(d-x)^2}{(d-x)^2 + h^2} = \frac{x^2}{x^2 + h^2}$$

$$(x^2 + h^2)(d-x)^2 = x^2((d-x)^2 + h^2)$$

(5)

REFRACTION

$$l^2 = h^2 + x^2$$

$$\sin \theta_i = \frac{x}{l} = \frac{x}{(h^2 + x^2)^{1/2}}$$

$$t = t_{SB} + t_{BR}$$

$$t = \frac{l_{SB}}{v_1} + \frac{l_{BR}}{v_2}$$

$$l_{SB} = (x^2 + h^2)^{1/2}$$

$$l_{BR} = ((d-x)^2 + h^2)^{1/2}$$

$$\min \left[t = \frac{(x^2 + h^2)^{1/2}}{v_1} + \frac{((d-x)^2 + h^2)^{1/2}}{v_2} \right]$$

$$\frac{dt}{dx} = 0 = \frac{1}{v_1} \left(\frac{x}{(h^2 + x^2)^{1/2}} \right) - \frac{1}{v_2} \left(\frac{d-x}{(h^2 + (d-x)^2)^{1/2}} \right)$$

AGAIN, TAKE DERIVATIVE
SET TO ZERO

$$= \frac{1}{v_1} \sin \theta_i - \frac{1}{v_2} \sin \theta_r$$

$$\frac{\sin \theta_r}{v_2} = \frac{\sin \theta_i}{v_1} \Rightarrow \text{SNELL'S LAW.}$$

CRITICAL ANGLE, θ_c & $\theta_r = 90^\circ$

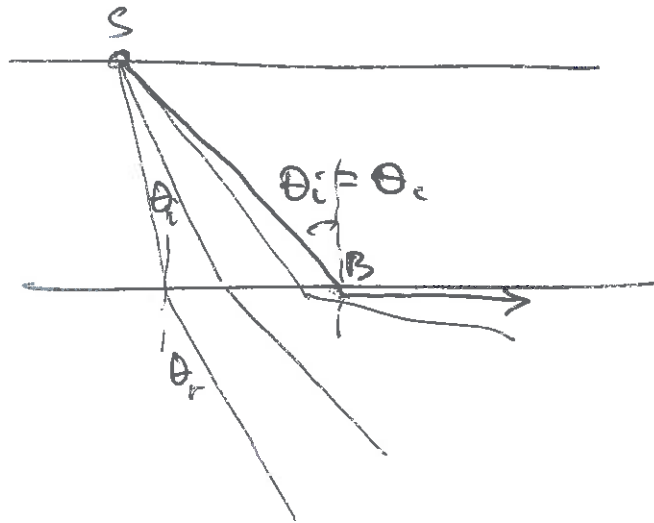
$$\frac{1}{v_2} = \frac{\sin \theta_c}{v_1}$$

$$\sin \theta_c = \frac{v_1}{v_2}$$

CRITICAL ANGLE, θ_c

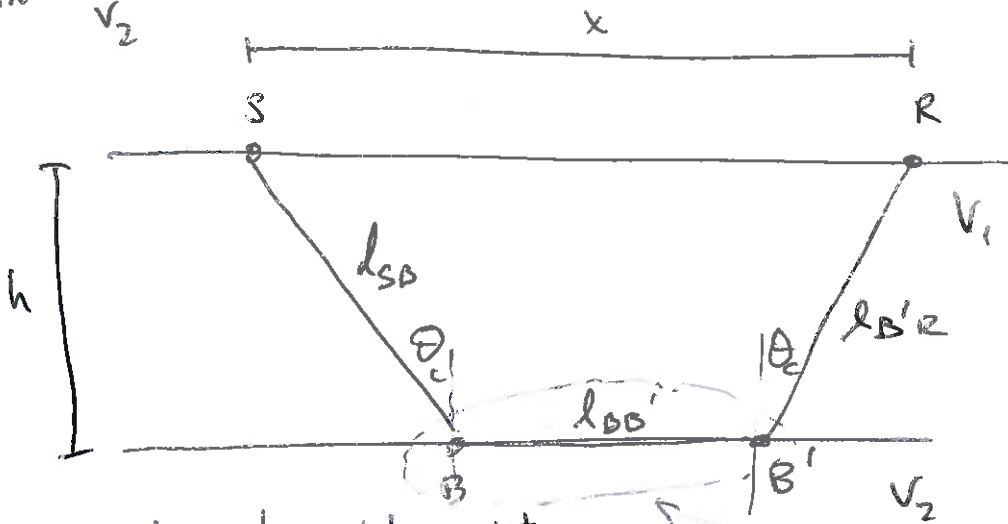
$$\theta_r = 90^\circ$$

$$\frac{1}{v_2} = \frac{\sin \theta_c}{v_1}$$



CRITICAL REFRACTION

$$\sin^{-1} \frac{v_1}{v_2} = \theta_c$$



$$t = t_{SB} + t_{BB'} + t_{B'R}$$

$$t = \frac{l_{SB}}{v_1} + \frac{l_{BB'}}{v_2} + \frac{l_{B'R}}{v_1}$$

WAVE TRAVELS JUST
BELOW INTERFACE
 $\Rightarrow v = v_2$

REFLECTION / TRANSMISSION COEFFICIENTS

NORMAL INCIDENT WAVE

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

↑
REFLECTION COEFFICIENT

← ACOUSTIC IMPEDANCE

$$Z = \rho v$$

↑
DENSITY ↑
VELOCITY

REFLECTION COEFFICIENT

$$R = \frac{\rho_2 v_2 - \rho_1 v_1}{\rho_2 v_2 + \rho_1 v_1}$$

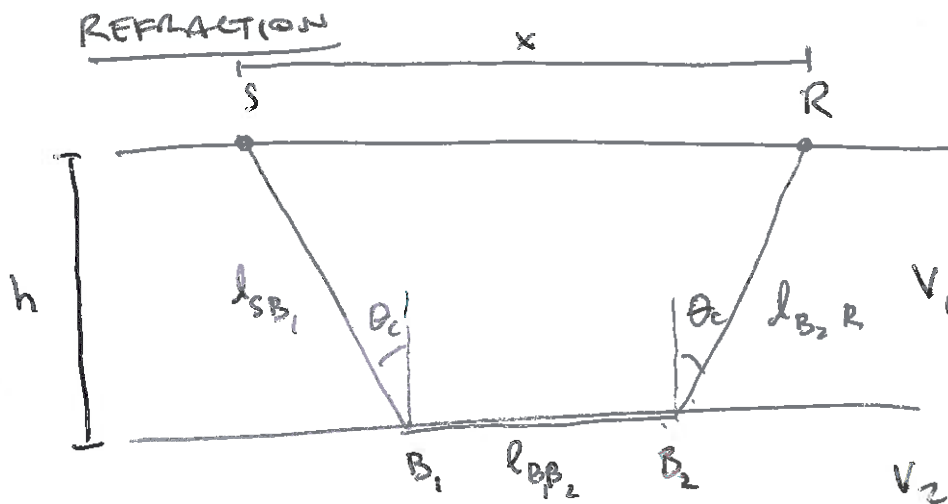
$R \rightarrow 1$ ALL ENERGY REFLECTED

$R \rightarrow 0$ ALL ENERGY IS TRANSMITTED

$$T = \frac{2Z_1}{Z_2 + Z_1}$$

↑
TRANSMISSION COEFFICIENT

TRANSMISSION COEFFICIENT



$$t = \frac{l_{SB_1}}{v_1} + \frac{l_{B_1B_2}}{v_2} + \frac{l_{B_2R}}{v_1}$$